

Theo Gerst

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EDUCATION

Massachusetts Institute of Technology, Cambridge, Massachusetts

Expected Graduation: May 2027

- **Relevant Coursework:** Fundamentals of Programming, Multivariable Calculus, C and Assembly, Linear Algebra, Discrete Math, Real Analysis, Algorithms, Abstract Algebra I, Theory of Computation, Computation Structures, Differential Equations, Abstract Algebra II, Functional Analysis, Topology, Software Construction, Circuits, Operating Systems, FPGAs, Performance Engineering, Systems, Database Systems.
- **Majors:** Pure Mathematics and Computer Science, **Cumulative Undergraduate GPA:** 5.0/5.0.

EXTRACURRICULARS

MIT Motorsports, Cambridge, Massachusetts

September 2023 – June 2026

Software Lead

- In 2024 season, wrote firmware for dashboard and custom inverters. Tested our telemetry system.
- In 2025 season, wrote and tested firmware for battery management system, vehicle control unit, dash, sensors, and telemetry.
 - The robust battery management and telemetry systems helped us complete endurance for the first time in 6 years.
- Use STM32s with embedded C++, RTOS, SPI/ISOSPI, I2C, CANFD, time series databases, asynchronous procedures.

MIT Arcturus, Cambridge, Massachusetts

September 2023 – May 2024

Autonomy Lead

- Designed and implemented controls system for autonomous boat. Also designed path-generating algorithm to navigate through cones, emergency stop system. Used Python and C++ with ROS2, RViz, Gazebo.

EXPERIENCES

Tesla, Inc., Palo Alto, California

May 2026 – August 2026

High Voltage Systems Firmware Intern

- Improved energy estimation algorithms, software-in-the-loop testing, and the interface between the BMS and the dashboard.

MIT Biomechatronics Lab, Cambridge, Massachusetts

June 2025 – August 2025

Undergraduate Research Assistant

- Wrote low-level drivers in C++ to run on a Toradex Verdin processor with Linux.
- Designed a custom power management PCB for a powered prosthetic knee, and a custom PCB with CAN transceivers, an ADC, and an IMU. Assembled and tested. Used KiCad and Altium.

Lawrence Berkeley Lab, AMCR Division, Berkeley, California

June 2024 – August 2024

Research Assistant

- Researched the behavior of various neuromorphic computation models. Learned factoring algorithms and designed an implementation for the General Number Field Sieve on a neuromorphic computer. Wrote a paper on the algorithm.

UC Berkeley Model Predictive Control Lab, Berkeley, California

June 2023 – August 2023

Intern

- Controlled 6-DOF robotic arm. Designed motion procedures, traced image contours on whiteboard.
- Ran custom simulations to predict robot behavior. Used Python and C++ with ROS, RViz, Gazebo.

PROJECTS

Radish, Berkeley, California

February 2022 – October 2022

Programming Language Developer

- Designed and implemented [Radish](#), a dynamically-typed programming language similar to JavaScript.
- Wrote lexical analyzer, parser, interpreter, an [IDE language extension](#) for autocomplete and highlighting, and a [standard library](#).
- Used C# for the language, Node JS for the language extension.

Discits, Berkeley, California

November 2020 – February 2022

Game Developer

- Designed and implemented [Discits](#), a large text-based RPG similar to Pokemon that runs via a messaging app.
- Implemented database collection system, battle system, map system. Used Node JS with mongoose for accessing database.

SKILLS

- **Programming Languages/Frameworks:** Python, C/C++, Node JS, C#, RISC-V Assembly, Java, Kotlin, Lisp, HTML/CSS/JS, Go, ROS(2), STM32, RTOS, Unreal Engine, Git, Bash, KiCad, Altium.